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 Studierichting: Informatica
 Oefeningenreeks 7B nr. 6.1

$$\begin{cases} f: \mathbb{R}^3 \rightarrow \mathbb{R} : (x, y, z) \rightarrow xyz \\ g: \mathbb{R} \rightarrow \mathbb{R}^2 : x \rightarrow (2x, x^3) \\ h: \mathbb{R}^2 \rightarrow \mathbb{R}^3 : (x, y) \rightarrow (x^2, y^2, x^2y) \end{cases}$$

$$f(x, y, z) = xyz \Rightarrow Df(x, y, z) = \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$g(x) = (2x, x^3) \Rightarrow Dg(x) = \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$h(x, y) = (x^2, y^2, x^2y) \Rightarrow Dh(x, y) = \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix}$$

$$\text{Deel 1: } D(h \circ g \circ f)(x, y, z)$$

$$D(h \circ g \circ f)(x, y, z) = Dh(g(f(x, y, z)))Dg(f(x, y, z))Df(x, y, z)$$

$$Dh(g(f(x, y, z))) = Dh(2xyz, x^3y^3z^3) = \begin{pmatrix} 4xyz & 0 \\ 0 & 2x^3y^3z^3 \\ 4x^4y^4z^4 & 4x^2y^2z^2 \end{pmatrix}$$

$$Dg(f(x, y, z)) = Dg(xyz) = \begin{pmatrix} 2 \\ 3x^2y^2z^2 \end{pmatrix}$$

$$Df(x, y, z) = \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$\Rightarrow D(h \circ g \circ f)(x, y, z) = \begin{pmatrix} 4xyz & 0 \\ 0 & 2x^3y^3z^3 \\ 4x^4y^4z^4 & 4x^2y^2z^2 \end{pmatrix} \begin{pmatrix} 2 \\ 3x^2y^2z^2 \end{pmatrix} \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$= \begin{pmatrix} 8xyz \\ 6x^5y^5z^5 \\ 8x^4y^4z^4 + 12x^4y^4z^4 \end{pmatrix} \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$= \begin{pmatrix} 8xyz \\ 6x^5y^5z^5 \\ 20x^4y^4z^4 \end{pmatrix} \begin{pmatrix} yz & xz & xy \end{pmatrix}$$

$$= \begin{pmatrix} 8xy^2z^2 & 8x^2yz^2 & 8x^2y^2z \\ 6x^5y^6z^6 & 6x^6y^5z^6 & 6x^6y^6z^5 \\ 20x^4y^5z^5 & 20x^5y^4z^5 & 20x^5y^5z^4 \end{pmatrix}$$

$$\text{Deel 2: } D(g \circ f \circ h)(x, y)$$

$$D(g \circ f \circ h)(x, y) = Dg(f(h(x, y)))Df(h(x, y))Dh(x, y)$$

$$Dg(f(h(x, y))) = Dg(x^4y^3) = \begin{pmatrix} 2 \\ 3x^8y^6 \end{pmatrix}$$

$$Df(h(x, y)) = Df(x^2, y^2, x^2y) = \begin{pmatrix} x^2y^3 & x^4y & x^2y^2 \end{pmatrix}$$

$$\begin{aligned}
Dh(x, y) &= \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix} \\
\Rightarrow D(g \circ f \circ h)(x, y) &= \begin{pmatrix} 2 \\ 3x^8y^6 \end{pmatrix} \begin{pmatrix} x^2y^3 & x^4y & x^2y^2 \end{pmatrix} \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix} \\
&= \begin{pmatrix} 2x^2y^3 & 2x^4y & 2x^2y^2 \\ 3x^{10}y^9 & 3x^{12}y^7 & 3x^{10}y^8 \end{pmatrix} \begin{pmatrix} 2x & 0 \\ 0 & 2y \\ 2xy & x^2 \end{pmatrix} \\
&= \begin{pmatrix} 4x^3y^3 + 4x^3y^3 & 4x^4y^2 + 2x^4y^2 \\ 6x^{11}y^9 + 6x^{11}y^9 & 6x^{12}y^8 + 3x^{12}y^8 \end{pmatrix} \\
&= \begin{pmatrix} 8x^3y^3 & 6x^4y^2 \\ 12x^{11}y^9 & 9x^{12}y^8 \end{pmatrix}
\end{aligned}$$

Deel 3: $D(f \circ h \circ g)(x)$

$$D(f \circ h \circ g)(x) = Df(h(g(x)))Dh(g(x))Dg(x)$$

$$Df(h(g(x))) = Df(4x^2, x^6, 4x^5) = \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix}$$

$$Dh(g(x)) = Dh(2x, x^3) = \begin{pmatrix} 4x & 0 \\ 0 & 2x^3 \\ 4x^4 & 4x^2 \end{pmatrix}$$

$$Dg(x) = \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$\Rightarrow D(f \circ h \circ g)(x) = \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix} \begin{pmatrix} 4x & 0 \\ 0 & 2x^3 \\ 4x^4 & 4x^2 \end{pmatrix} \begin{pmatrix} 2 \\ 3x^2 \end{pmatrix}$$

$$= \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix} \begin{pmatrix} 8x \\ 6x^5 \\ 8x^4 + 12x^4 \end{pmatrix}$$

$$= \begin{pmatrix} 4x^{11} & 16x^7 & 4x^8 \end{pmatrix} \begin{pmatrix} 8x \\ 6x^5 \\ 20x^4 \end{pmatrix}$$

$$= 32x^{12} + 96x^{12} + 80x^{12}$$

$$= 208x^{12}$$

References